

4. (25 marks) Consider the following quadratic form

$$Q(\mathbf{x}) = abx_1^2 + bx_2^2 + ax_3^2 + 2bx_2x_3.$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$$

Find all the principal submatrices of A .

Solutions: (2 marks) The matrix of coefficients is:

$$A = \begin{pmatrix} ab & 0 & 0 \\ 0 & b & b \\ 0 & b & a \end{pmatrix}$$

(2 marks) Its principal submatrices are:

$$\begin{aligned} &3^{\text{rd}} \text{ Order: } \begin{pmatrix} ab & 0 & 0 \\ 0 & b & b \\ 0 & b & a \end{pmatrix} \\ &2^{\text{nd}} \text{ Order: } \begin{pmatrix} ab & 0 \\ 0 & b \end{pmatrix}, \begin{pmatrix} ab & 0 \\ 0 & a \end{pmatrix} \text{ and } \begin{pmatrix} b & b \\ b & a \end{pmatrix}, \\ &1^{\text{st}} \text{ Order: } (ab), (b) \text{ and } (a). \end{aligned}$$

- (b) (6 marks) Define what it means for the quadratic form $Q(\mathbf{x})$ to be *negative definite*. How would you be able to assess, in terms of the properties of A , whether $Q(\mathbf{x})$ is negative definite?

Solutions: (3 marks) $Q(\mathbf{x})$ is negative definite if $Q(\mathbf{x}) < 0$ for any $\mathbf{x} \in \mathbb{R}^3 \setminus \mathbf{0}$.

(3 marks) To, check if $Q(\mathbf{x})$ is negative we need to verify that all odd leading principal minors are negative and that all even leading principal minors are positive.

- (c) (15 marks) Find the values of the parameter values, a and b , such that the quadratic form $Q(\mathbf{x})$ is respectively *indefinite*, *positive definite* and *negative definite*? Then, plot these regions in $\mathbb{R}^2$.

Solutions: (9 Marks) $Q(\mathbf{x})$ is positive definite if all leading principal minors are positive which requires:

$$ab > 0, \quad ab^2 > 0, \quad ab^2(a - b) > 0.$$

For these to hold at once it must be that

$$a > b > 0.$$

$Q(\mathbf{x})$ is negative definite is all leading principal minors suitably alternate in sign which requires:

$$ab < 0, \quad ab^2 > 0, \quad ab^2(a - b) < 0.$$

But these conditions never hold at once as the first two require $a > 0$ and $b < 0$. But if so, $ab^2(a - b) > 0$.

For checking indefiniteness we need to rule out that the matrix is semidefinite. For $Q(\mathbf{x})$ to be positive semidefinite, we need all principal minor to be non-negative, or equivalently

$$a \geq b \geq 0.$$

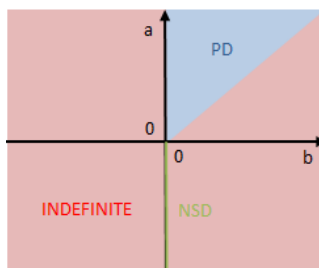
For $Q(\mathbf{x})$ to be negative semidefinite, we need all principal minor to suitably alternate in signs. By looking at first order principal minors, we have that

$$ab \leq 0, \quad a \leq 0, \quad b \leq 0.$$

For these to hold we need that either $a = 0$ and $b < 0$ or that $b = 0$ and $a \leq 0$. But the former case cannot occur as a second order principal minor would be negative as $ab - b^2 < 0$. Thus, the matrix is negative semidefinite only when $b = 0$ and $a \leq 0$. Thus, the quadratic form is indefinite if and only if one

$$\text{either } b < 0 \text{ or } b > 0 \text{ and } a < b.$$

The plot below depicts the answer



5. (25 marks) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a *differentiable convex function*.

- (a) (6 marks) State the definitions of a *critical point* and of a *global minimizer* for the function f .

Solutions: (3 marks) A critical point x of the function f is a point at which $f'(x) = 0$.

(2 marks) A global minimizer x of the function f is a point at which $f(x) \leq f(y)$ for any $y \in \mathbb{R}$.

- (b) (10 marks) State definition of *convex function* for the function f . Then, construct an equivalent definition of *convexity for differentiable function* f .

Solutions: (3 marks) The function f is convex if for any $x, y \in \mathbb{R}$

$$f(ty + (1 - t)x) \leq tf(y) + (1 - t)f(x) \quad \text{for all } t \in [0, 1].$$

(4 marks) The differentiable function f is convex if for any $x, y \in \mathbb{R}$

$$(y - x)f'(x) \leq f(y) - f(x).$$

(3 marks) To show that the two definitions are equivalent note that

$$\begin{aligned} tf(y) + (1 - t)f(x) \geq f(ty + (1 - t)x) &\Leftrightarrow t(f(y) - f(x)) + f(x) \geq f(x + t(y - x)) \Leftrightarrow \\ f(y) - f(x) &\geq \frac{f(x + t(y - x)) - f(x)}{t} \Leftrightarrow f(y) - f(x) \geq \frac{f(x + h) - f(x)}{h}(y - x), \end{aligned}$$

for $h = t(y - x)$. Taking limits when $h \rightarrow 0$, the latter implies that

$$f(y) - f(x) \geq f'(x)(y - x). \quad (1)$$

To prove the converse, let $z = ty + (1 - t)x$. By applying equation (1), we have that

$$\begin{aligned} f(y) - f(z) &\geq f'(z)(y - z), \\ f(x) - f(z) &\geq f'(z)(x - z). \end{aligned}$$

Summing the first inequality multiplied by t to the second multiplied by $(1 - t)$, we find the result

$$[f(y) - f(z)]t + [f(x) - f(z)](1 - t) \geq 0 \Leftrightarrow tf(y) + (1 - t)f(x) \geq f(z).$$

- (c) (9 marks) By exploiting the previous definition, show that any *critical point* of f is a *global minimizer* of f .

Solutions: (9 marks) If x is a critical point of f , by the second definition of convexity we get that for any $y \in \mathbb{R}$

$$0 = (y - x)f'(x) \leq f(y) - f(x) \Rightarrow f(x) \leq f(y).$$

Thus, x is a minimizer.

6. (50 marks) For $a > 0$, consider the problem:

$$\begin{aligned} \max_{x,y} \quad & ax + \log y \\ \text{s.t.} \quad & 2x + 2y \leq 7 \\ & y \geq a, x \geq 0. \end{aligned}$$

- (a) (8 marks) Establish whether the objective function and the constraints are respectively concave, convex, or neither. Can you solve this optimization problem using the Kuhn-Tucker method? That is, for what values of $a > 0$ is the Slater condition satisfied?

Solutions: (2 marks) The constraints are obviously linear and thus both convex and concave. (2 marks) The objective function is concave, as its Hessian is negative semidefinite for $y \geq a > 0$,

$$D^2 f(x, y) = \begin{pmatrix} 0 & 0 \\ 0 & -1/y^2 \end{pmatrix}, \quad |D^2 F(x, y)| = 0 \geq 0, \quad -1/y^2 < 0 \text{ and } 0 \leq 0.$$

(4 marks) When the constraint set has a non-empty interior, we can adopt the Kuhn-Tucker approach to solve the problem. This is the case for all $a < 7/2$ since the point $\tilde{x} = \varepsilon$ and $\tilde{y} = a + \varepsilon$ is in the interior of the constraint set for $\varepsilon < (7 - 2a)/4$, as

$$2\tilde{x} + 2\tilde{y} = 2a + 4\varepsilon < 7, \quad \tilde{x} > 0, \quad \tilde{y} > a.$$

- (b) (8 marks) Suppose Slater condition is satisfied, write down the Kuhn-Tucker necessary first order conditions that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangean?

Solutions: (6 marks) The necessary first order conditions for the solution of the constrained maximization problem are:

$$\begin{aligned} 0 &= a - 2\lambda_1 + \lambda_2, \\ 0 &= 1/y - 2\lambda_1 + \lambda_3, \\ 0 &= \lambda_1(2x + 2y - 7), \\ 0 &= \lambda_2 x = \lambda_3(y - a), \\ 0 &\geq 2x + 2y - 7, \quad 0 \geq -x, \quad 0 \geq a - y, \\ 0 &\leq \lambda_1, \quad 0 \leq \lambda_2, \quad 0 \leq \lambda_3. \end{aligned}$$

(2 marks) Any solutions to these conditions is maximizer of the Lagrangean as the Lagrangean is a sum of concave functions by the arguments in (a) and (b).

- (c) (20 marks) Suppose Slater condition is satisfied, solve the constrained optimization problem in terms of a .

Solutions: Consider the various supports:

- (2 marks) At an optimum it also cannot be that $\lambda_1 = 0$, as the FOC for good x would imply that, $\lambda_2 = -a < 0$. So there is no solution on this support.
- (2 marks) At an optimum it also cannot be that $\lambda_1, \lambda_2, \lambda_3 > 0$ as it would imply $x = 0$, $y = a$ and $2x + 2y = 2a = 7$ which cannot if Slater condition holds.
- (5 marks) Next consider the case in which $\lambda_1, \lambda_2 > 0$ while $\lambda_3 = 0$. If so, the two binding constraints would imply that

$$y(a) = \frac{7}{2} \geq a \quad \text{and} \quad x(a) = 0.$$

Solving the two FOC for goods x and y would imply that

$$\lambda_1(a) = \frac{1}{7} \geq 0, \quad \lambda_2(a) = \frac{2}{7} - a \geq 0.$$

But then $\lambda_2(a) \geq 0$ would require $a \leq 2/7$. Thus, there is a solution on this support if and only if $a \leq 2/7$.

- (5 marks) Next consider the case in which $\lambda_1, \lambda_3 > 0$ while $\lambda_2 = 0$. If so, the two binding constraints would imply that

$$y(a) = a \quad \text{and} \quad x(a) = \frac{7}{2} - a.$$

Solving the two FOC for goods x and y would imply that

$$\lambda_1(a) = \frac{a}{2} \geq 0, \quad \lambda_3(a) = a - \frac{1}{a} \geq 0.$$

But then $\lambda_3(a) \geq 0$ would require $a \geq 1$. Thus, there is a solution on this support if and only if $a \geq 1$.

- (6 marks) In the final scenario, the optimum must live in the only remaining support, namely $\lambda_1 > 0$ and $\lambda_3 = \lambda_2 = 0$. Solving FOC on the binding constraint at such values one finds that

$$y(a) = \frac{1}{a}, \quad x(a) = \frac{7}{2} - \frac{1}{a}, \quad \lambda_1(a) = \frac{a}{2} > 0.$$

For this to be a solution it must be that

$$y(a) \geq a \quad \& \quad x(a) \geq 0 \quad \Rightarrow \quad a \in [2/7, 1].$$

Thus, there is a solution on this support if and only if $a \in [2/7, 1]$.

- (d) (10 marks) State Envelope Theorem and apply to find the slope of value function with respect to the parameter a for $a \in [0, 3]$.

Solutions: (4 marks) The Envelope Theorem in general states that if $(\mathbf{x}(a), \boldsymbol{\lambda}(a))$ are differentiable functions and that $\mathbf{x}(a)$ satisfies the constraint qualification. If so,

$$\frac{dv(a)}{da} = \frac{\partial L(\mathbf{x}(a), \boldsymbol{\lambda}(a)|a)}{\partial a} = x(a) - \lambda_3(a)$$

(2 marks) If $a \leq 2/7$ and thus $\lambda_3(a) = 0$, value is independent of a

$$\frac{dv(a)}{da} = x(a) = 0.$$

(2 marks) If $a \in (2/7, 1]$ and thus $\lambda_3(a) = 0$, we have that the value increases in a

$$\frac{dv(a)}{da} = x(a) = \frac{7}{2} - \frac{1}{a} > 0.$$

(2 marks) If $a \geq 1$ and thus $\lambda_3(a) > 0$, the slope of the value with respect to a amounts to

$$\frac{dv(a)}{da} = x(a) - \lambda_3(a) = \frac{7}{2} + \frac{1}{a} - 2a.$$

(e) (4 marks) For what value of $a \in [0, 3]$ is the value function maximized?

Solutions: To maximize the value with respect to a , we solve

$$\max_{a \in [1, 3]} v(a).$$

As $dv(a)/da = 0$ for $a = 2$ and $d^2v(a)/(da)^2 < 0$ for all $a \in [1, 3]$, the solution must set $a = 2$.